



# Field larvicidal and oviposition repellent efficacy of three essential oil formulations for the control of filaria vector mosquito *Culex quinquefasciatus* Say (Diptera: Culicidae): A kill and push concept

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## ABSTRACT

Biological control of larval mosquitoes is in great demand due to the development of resistance against synthetic insecticides, environmental toxicity and the inability to protect habitats from further oviposition. In the present study, three botanical essential oils (BEOs) – citronella, eucalyptus, and pine oils – were formulated for the assessment of larvicidal and oviposition repellent efficacies against *Culex quinquefasciatus* Say, the filaria vector. The GC–MS profiling of BEOs showed the presence of 16 – 19 compounds covering 87.7–93% of oil composition. The resistance status of *Culex quinquefasciatus* population was evaluated with temephos (LC<sub>50</sub> = 0.001 ppm, LC<sub>90</sub> = 0.01 ppm). Larval bioassay of emulsifiable concentrate (EC) formulations prepared from eucalyptus and pine oils showed promising efficacy (LC<sub>50</sub> = 22.7 and 23.2 ppm) and LC<sub>90</sub> (63.8 and 62.4 ppm) compared to citronella oil EC (LC<sub>50</sub> = 43.4 ppm and LC<sub>90</sub> = 199.0 ppm). The field trials of eucalyptus + pine (1:1 ratio) EC showed 100% larval mortality for 3 weeks at 300 ppm compared to 2 weeks of individual oils. Further, the oviposition attraction index (OAI) for ECs of eucalyptus, pine, and their combination showed complete protection of breeding habitats from oviposition at 1st week and –0.9 to –1.0 OAI at 2nd week with slight reduction to –0.5 at 3rd week. Citronella EC provided shortest larvicidal and oviposition repellent efficacy under the field conditions. The promising mosquitocidal activities of EC formulations of eucalyptus and pine or their combination suggest them as potential biocontrol vector control candidates over citronella oil.

## Introduction

Mosquitoes are the important vectors of pathogenic protozoans, nematodes and viruses affecting 500 million individuals every year causing malaria, lymphatic filaria, chikungunya, dengue, Japanese encephalitis, West Nile and Zika diseases (CDC, 2021; WHO, 2022). Among these VBDs, filaria is mainly transmitted by *Culex quinquefasciatus* mosquitoes which produced several types of disability that have known limited treatment options (Reuben et al., 1994; National Center for Vector Borne Diseases, 2022). In India, National Filaria Control Programme (NFCP) was launched in 1955 with a set the goal for elimination by 2015 which is now extended to 2030 by Ministry of Health and

Family Welfare, Govt. of India (<https://www.pib.gov.in/>). The enhancement in vector management tactics is essential to manage mosquito-borne disease transmission effectively.

Till date, vector controls centrally rely on the use of various synthetic insecticides against adult and larval stages (WHO, 2009; Fonseca et al., 2013; Unlu et al., 2017). Although they are effective and cheaper, frequent repetitive applications are required to keep mosquito density low for a season causing significant environmental toxicity and insecticide resistance (Fonseca et al., 2013; Suman et al., 2014; Sun et al., 2014; Farajollahi et al., 2013; Williams et al., 2014). The development of insecticide resistance compromises the success of vector control programme (Brogdon and McAllister, 1998; Marcombe et al., 2014). Thus,

**Abbreviations:** LC, Lethal concentration; CN, Citronella; EU, Eucalyptus; PN, Pine.

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the finding of new agents with multiple modes of action may help in the development of a better vector control strategy.

Botanical essential oils (BEOs) are the great biological resources for eco-friendly mosquito management which have been evaluated against various mosquito species (Amer and Mehlhorn, 2006; Ansari et al., 2005; Araujo et al., 2016; Benelli et al., 2015; Govindarajan et al., 2013; Maia and Moore, 2011; Mdoe et al., 2014; Shaalan et al., 2005; Suman et al., 2010). Essential oils contain numerous compounds that have shown larvicidal activities against several mosquito species (Andrade-Ochoa et al., 2018; Cheng et al., 2009; Ghosh et al., 2012). However, the efficacy of essential oils has been undermined as the majority of studies considered plant extracts and unformulated BEOs in solvent extracts (Ghosh et al., 2012). Sugumar et al. (2014) developed nanoemulsion of eucalyptus oil to enhance larvicidal activity against *Cx. quinquefasciatus*, but 200–250 ppm was required to achieve higher mortality. Therefore, the development of an efficient formulation of botanical oils may enhance larvicidal efficacy and provide a promising eco-friendly tool.

Moreover, several essential oils have also been found helpful in insecticide resistance maintenance which were effective to enhance larvicidal activity in various mosquito populations resistant to different insecticides (Elia-Amira et al., 2018; Gross et al., 2017; Norris et al., 2019). This suggests great potential of BEOs as an alternative or combination treatment for sustainable control of mosquitoes.

Recently, explorations on oviposition repellents have been encouraged to reduce mosquito biting pressure. Fewer studies are reported on mosquito oviposition repellent or deterrent activities of BEOs (Benelli, 2015; Coria et al., 2008; Kempraj and Bhat, 2011; Maheswaran and Ignacimuthu, 2012; Xue et al., 2001). Tikar et al. (2014) have shown the presence of oviposition repellency in synthetic adult repellents such as diethyl phenyl acetamide (DEPA), N, N-diethyl-meta-toluamide (DEET), and N,N-diethylbenzamide (DEB). Essential oils have been used for biting protection against blood-seeking mosquitoes (Ansari et al., 2005; Trigg, 1996a; Trigg, 1996b; Kanis et al., 2009; Hsu et al., 2013). These studies also signify the use of repellents to inhibit oviposition in treated mosquito breeding habitats. The current findings on oviposition deterrence of essential oil components, (E)-Caryophyllene and  $\alpha$ -Humulene, against *Aedes aegypti* by da Silva et al. (2015) suggest the need for further explorations of various essential oils for the protection of breeding habitats. There is a lack of enough evidence on possession of long-term larviciding and oviposition repellency in a single formulation under field conditions.

By considering the above facts, to find such dual biological activity, three BEOs, i.e., citronella, eucalyptus and pine oils were assessed to develop emulsifiable formulations (EC) to evaluate larvicidal and oviposition repellent activities under laboratory and field conditions. The baseline larvicidal efficacy was estimated against the filarial vector, *Cx. quinquefasciatus* under laboratory conditions. The long-term small-scale field trials were conducted for larvicide and oviposition repellency in *peri*-domestic areas. These biological formulations may also help in the management of insecticide resistance in container mosquitoes.

## Materials and methods

### Mosquito collection and rearing

The first and second instar larvae of *Cx. quinquefasciatus* were collected from natural breeding places, i.e., drains, ponds and large containers from Kolkata, West Bengal, India. Larvae were brought to the laboratory at Zoological Survey of India, Kolkata, West Bengal, India for further rearing to 3rd – early 4th instars under  $26 \pm 1$  °C and 12 h photoperiods (Suman et al., 2010). Larvae were fed on dried Brewer's yeast (0.3 gm/L). The larval rearing water was changed on alternate days with dechlorinated tap water. The adult females were blood-fed on a membrane feeding device having temperature control (37 °C) with a hot water supply. The blood was arranged from certified goat slaughterhouses in EDTA treated tubes to avoid clotting.

### Ethical statement

No live animal was used for mosquito rearing and no mosquito was released in the field for any experiment. The blood provider slaughterhouse was informed and asked for consent to use the blood for mosquito feeding. The study protocol was approved by the institutional ethical committee.

### Botanical essential oils (BEOs) and chemical composition

The three crude BEOs, citronella (*Cymbopogon nardus* (L.) Rendle), eucalyptus (*Eucalyptus citriodora* Hook), and pine (*Pinus roxburghii* Sarg.) oils were obtained from a local commercial supplier (BioGuard Eco Solutions Inc., Kolkata (WB), India). These oils were sent to CEG Test House and Research Centre Pvt. Ltd., Jaipur (RJ), India to find their chemical composition using gas chromatography-mass spectroscopy (GC–MS) method on Thermo GC Trace 1300 instrument with GC–MS/MS TSQ8000 detector and DB – 5MS (30 m  $\times$  0.25 mm  $\times$  0.25  $\mu$ m) column. GC–MS samples were prepared in *n*-hexane (pine and eucalyptus oils) and acetonitrile (citronella oil). The ion source and interface temperatures were 250 °C and 280 °C, respectively. The total run time was 51 min. to generate a chromatogram. The chromatogram peak and area profiles were matched with the reference chemical library for the detection of chemicals. The retention time (RT), peak area and peak height were used to estimate the proportion of the chemical compounds in the essential oils. The instruments were calibrated before each run.

### Susceptibility status of larvae *Culex* mosquitoes for temephos

A sample of temephos (95.1%) technical compound “(O-[4-((Dimethoxyphosphorothioyl) oxy) phenyl] sulfanyl) phenyl] O, O-dimethyl phosphorothioate)” (CAS # 3383-96-8) was gifted by Hernaba Industries, GLDC, Vapi (Gujarat), India. A 1% stock solution of temephos was prepared in acetone and supplemented with 5% Tween – 20 emulsifying agents. To assess the susceptibility level of mosquito larvae, 20 late 3rd – early 4th instars of *Cx. quinquefasciatus* were exposed to 4–5 doses between  $5.0 \times 10^{-3}$  to 0.1 ppm of temephos in 100 ml tap water filled in polypropylene plastic cups for the laboratory bioassay (WHO, 2005). Control cups were treated with 1 ml acetone 30 min. prior to the releasing of larvae. After 24 h exposure, observations were made on live and dead larvae to estimate mortality for various doses of temephos. The bioassay was conducted in three replicates and repeated three times.

### Development of emulsifiable concentrate (EC) formulations and larvicidal evaluations

Various EC formulations of citronella, eucalyptus, and pine oils were prepared for the treatment of bioassay containers and larval habitats in the aqueous medium. These EC formulations were comprised of 50% (V/V) BEO (citronella, eucalyptus or pine oils) as alone or in combinations of eucalyptus and pine (1:1 ratio) with other 50% components including soya refined oil as a carrier (39%), Tween-20, Tween – 60 and Span – 20 as surfactants (10%) and stabilizers and Butylated hydroxyanisole (BHA) + Butylated hydroxytoluene (BHT) as anti-oxidants (1%). These formulations were optimized for the phase separation, droplet size, and distribution during storage and dilutions (data not shown).

For larval susceptibility bioassay under the laboratory conditions, fresh 1% stock solutions of different BEOs were prepared and serially diluted in distilled water (10 – 160 ppm) for the treatment of larvae in the bioassay containers. Third instar larvae of *Cx. quinquefasciatus* were evaluated against these ECs as described above in larvicidal susceptibility for temephos. The control was the blend of all the compounds except active essential oil. The experiment was conducted in three replicates and repeated at least three times.

### Larvicidal efficacy of essential oil formulations under field conditions

The field larvicidal efficacy of EC formulations containing citronella, eucalyptus, pine oils individually, and a combination of eucalyptus and pine (1:1 ratio) oils were estimated against *Cx. quinquefasciatus* larvae in 20 L pond water-filled plastic buckets (30 L). For bioassay, healthy 3rd instars *Cx. quinquefasciatus* larvae were selected and transferred into larval trays for 2–3 h to acclimatize to the field conditions and were offered dried Brewer's yeast for feeding (Suman et al., 2010). A 10% stock solution for each EC (citronella, eucalyptus, pine, and combination of eucalyptus and pine) was prepared in distilled water to achieve 300 ppm concentration in 20 L water in buckets. Each control and treatment bucket had received 100 third instar *Cx. quinquefasciatus* weekly. Control buckets were not treated with any chemicals as we did not receive any significant mortality in the blend in laboratory experiments. The buckets were covered with perforated polyethylene plastic film fitted at a 15–20° slanting angle on the bucket lid to avoid drop-in of plant leaves or dust. The plastic films were provisioned with 2 mm diam. holes ( $n = 100$ ) for proper aeration inside the bucket. These buckets were placed under the trees or shady areas to avoid exposure to direct bright sunlight and heat. Observations on dead or live larvae were carried out two times a week from all buckets of control and treatment. The emerged mosquitoes in both control and treatments buckets were aspirated and used to estimate the efficacy. New batches of 100 larvae were transferred weekly in the same buckets for 5 weeks, consecutively. The bioassay was conducted with three replicates and repeated twice.

### Oviposition repellency of emulsifiable formulations under field conditions

Oviposition repellency of citronella, eucalyptus, and pine oils individually, and a combination of eucalyptus and pine (1:1 ratio) oils in EC formulations were evaluated against *Cx. quinquefasciatus* under the field conditions. The study was conducted in the backyard of human dwellings having a high population density of *Cx. quinquefasciatus*. The presence of mosquitoes was confirmed using BG-Sentinel traps for 12 h in the night from 6 PM to 6 AM before deployment of ovitraps in the yards that showed 65–93 mosquito/trap. Ovitrap were prepared from brown plastic flower pots (1.5 L) and filled with 1 L pond water. All treatment ovitraps were treated with 300 ppm of oil formulations, but control ovitraps were not treated with any chemical. These ovitraps were placed in shady areas under the bushes and the trees to provide suitable habitat conditions. Eggrasts from both traps were collected weekly for the estimation of density as eggs per trap as described by Suman et al. (2011). The quantity of water evaporated during the experiment was replenished with the same pond water. The experiment was terminated after 3 weeks due to heavy rainfall that over flooded the ovitraps.

Oviposition activity index (OAI) was calculated using the formula described by Kramer and Mulla (1979):

$$OAI = (NT - NC) / (NT + NC)$$

Whereas, NT is the total number of eggs in the treatment ovitraps and NC is the total number of eggs in the control ovitraps. The OAI response ranges between +1 (maximum attractant) and –1 (maximum repellent). The OAI with +0.3 and above are considered as attractants while those with –0.3 and below are considered as repellents (Kramer and Mulla, 1979).

### Statistical analysis

The lethal concentrations, i.e., LC<sub>50</sub> and LC<sub>90</sub> with upper and lower fiducial limits at 95% level were estimated for temephos, crude, and EC formulations of citronella, eucalyptus, and pine oils using log - probit analysis (Finney, 1971). The data from various larvicidal bioassays were pooled dose-wise for all replicates and repeats. In the log - probit analysis, larvicidal doses and mortality were transformed using log (10) and

probit scale, respectively. A Chi-square test was conducted to estimate the heterogeneity with the degree of freedom ( $df$ ) and probability test ( $P$ ) values for temephos, crude oils and EC formulated BEOs against different mosquito species. Regression equations for dose–response slopes of different larvicides were prepared. The mortality in control was below 5%; hence, no correction factor was used to adjust the treatment mortality. One-way analysis of variance (ANOVA) was used to estimate significant differences among the efficacy of BEOs at different weeks for field larvicidal evaluations and oviposition repellency at  $p < 0.05$ . The least significant difference (LSD) method was used to compare the means in ANOVA for significant differences. PASW ver. 18 statistical packages were used for the analysis.

## Results

### Chemical composition of botanical essential oils

Chemical profiling of different essential oils was shown in Tables 1–3. Citronella oil showed the presence of 17 compounds comprising 88.0% of the oil. The major constituents were citronellol (19.6%), citronellal (18.6%), geraniol (11.6%), and citral (9.2%) (Table 1). In eucalyptus oil, 16 compounds (87.7%) were detected which were mainly 3-carene (14.1%), eucalyptol (12.3%), o-cymene (12.0%), and longifolene (8.0%) (Table 2). In pine oil, 19 compounds covered 93.0% of the oil composition mainly with  $\alpha$ -terpineol (15.3%), 3-carene (11.5%), (+)-4-carene (9.6%),  $\zeta$ -terpinene (8.9%), and longifolene (7.6%) (Table 3).

### Susceptibility of *Culex quinquefasciatus* larvae to temephos

Temephos showed considerable efficacy against larval *Cx. quinquefasciatus* mosquitoes in laboratory bioassays (Table 4). LC<sub>50</sub> of temephos was  $1.0 \times 10^{-3}$  ppm and LC<sub>90</sub> was  $1.0 \times 10^{-2}$  ppm for *Cx. quinquefasciatus* larval population used for assessment of the BEOs. No mortality in control was found.

### Estimation of baseline susceptibility doses of emulsifiable concentrate (EC) formulations against larval mosquitoes

The data showed a considerable larvicidal efficacy of EC formulations of citronella, eucalyptus, and pine oils against 3rd – 4th instars of *Culex* mosquitoes (Table 4). The mortality in controls was less than 5%.

Both LC<sub>50</sub> and LC<sub>90</sub> doses for *Cx. quinquefasciatus* were 43.4 and 199.0 ppm for citronella oil, respectively which were considerably

**Table 1**

Estimation of the chemical composition of citronella (*Cymbopogon nardus* (L.) Rendle) oil with gas chromatography-mass spectroscopy (GC–MS) method.

| Chemical compounds         | %           | RT   |
|----------------------------|-------------|------|
| 5-Hepten-2-one, 6-methyl-  | 1.3         | 9.6  |
| $\alpha$ -Murolene         | 0.6         | 24.6 |
| Caryophyllene              | 5.2         | 21.9 |
| Caryophyllene oxide        | 1.9         | 26.4 |
| Citral                     | 9.2         | 18.2 |
| Citronellal                | 18.6        | 14.5 |
| Citronellol                | 19.6        | 17.2 |
| Citronellyl acetate        | 4.5         | 20.0 |
| Decanal                    | 1.2         | 15.9 |
| Eugenol                    | 1.0         | 20.1 |
| Geraniol                   | 11.6        | 17.9 |
| Geranyl acetate            | 2.0         | 20.8 |
| Humulene                   | 0.9         | 22.8 |
| Linalool                   | 2.2         | 12.8 |
| Naphthalene                | 1.4         | 21.1 |
| trans-Verbenol             | 2.7         | 15.2 |
| Trifluoroacetyl-lavandulol | 4.0         | 17.9 |
| <b>Total</b>               | <b>88.0</b> |      |

RT – Retention time, % - Percentage of the chemical compound in the oil.

**Table 2**

Estimation of the chemical composition of eucalyptus (*Eucalyptus citriodora* Hook) oil with gas chromatography-mass spectroscopy (GC-MS) method.

| Chemical compounds                           | %           | RT   |
|----------------------------------------------|-------------|------|
| (-)-trans-Pinane                             | 5.5         | 7.3  |
| (1R)-2,6,6-Trimethylbicyclo[3.1.1]hept-2-ene | 3.2         | 7.7  |
| (-)-Car-3-en-2-one                           | 1.0         | 18.9 |
| (-)- $\alpha$ -thujene                       | 2.2         | 9.0  |
| (+)-4-Carene                                 | 5.3         | 12.4 |
| 3-Carene                                     | 14.1        | 10.0 |
| $\alpha$ -Pinene                             | 5.5         | 7.8  |
| Camphene, (1R,4S)-(+)-                       | 4.2         | 9.2  |
| $\gamma$ -Terpinene                          | 4.7         | 10.2 |
| Eucalyptol                                   | 12.3        | 10.8 |
| L- $\alpha$ -Terpineol                       | 3.7         | 15.7 |
| Longifolene                                  | 8.0         | 21.7 |
| Longifolene-(V4)                             | 2.3         | 20.7 |
| m-Cymen-8-ol                                 | 2.2         | 15.5 |
| o-Cymene                                     | 12.0        | 10.7 |
| Terpinen-4-ol                                | 1.2         | 15.2 |
| <b>Total</b>                                 | <b>87.5</b> |      |

RT – Retention time, % - Percentage of the chemical compound in the oil.

**Table 3**

Estimation of chemical composition of pine (*Pinus roxburghii* Sarg.) oil with gas chromatography-mass spectroscopy (GC-MS) method.

| Chemical compounds                           | %           | RT   |
|----------------------------------------------|-------------|------|
| (+)-2-Carene                                 | 3.8         | 10.5 |
| (+)-4-Carene                                 | 9.6         | 12.5 |
| 3-Carene                                     | 11.5        | 10.1 |
| 3-Cyclohexen-1-ol,1-methyl-4-(1-ethylethyl)- | 4.1         | 14.0 |
| $\alpha$ -Pinene                             | 6.0         | 7.8  |
| $\alpha$ -Terpineol                          | 15.3        | 16.0 |
| $\alpha$ -Terpinyl acetate                   | 2.4         | 16.0 |
| Camphene                                     | 3.0         | 8.3  |
| Caryophyllene                                | 0.9         | 22.1 |
| $\gamma$ -Terpinene                          | 8.9         | 10.4 |
| Cyclohexanol, 1-methyl-4-(1-methylethenyl)-  | 1.7         | 14.3 |
| Eucalyptol                                   | 2.1         | 10.9 |
| Fenchol                                      | 1.2         | 13.4 |
| Isoborneol                                   | 1.3         | 15.0 |
| Limonene                                     | 6.9         | 10.9 |
| Longifolene                                  | 7.6         | 21.7 |
| Longifolene-(V4)                             | 1.8         | 20.7 |
| o-Cymene                                     | 3.5         | 10.7 |
| Terpinen-4-ol                                | 1.4         | 15.2 |
| <b>Total</b>                                 | <b>93.0</b> |      |

RT – Retention time, % - Percentage of the chemical compound in the oil.

**Table 4**

Estimation of the susceptibility status of 3rd instars larvae of *Culex quinquefasciatus* against temephos organophosphate insecticide using Log (dose) – Probit (mortality) analysis. All the doses and fiducial limits (CI 95%) for LC<sub>50</sub> and LC<sub>90</sub> are in ppm.

| Larvicide      | LC <sub>50</sub> (CI 95%)<br>(in ppm)                                  | LC <sub>90</sub> (CI 95%)<br>(in ppm)                                   | Regression equation<br>(y = a + bx) | $\chi^2$<br>(df) | P-value |
|----------------|------------------------------------------------------------------------|-------------------------------------------------------------------------|-------------------------------------|------------------|---------|
| Temephos       | 1.0x10 <sup>-3</sup><br>(0.5x10 <sup>-3</sup> – 3.0x10 <sup>-3</sup> ) | 1.0x10 <sup>-2</sup><br>(0.4x10 <sup>-2</sup> – 17.2x10 <sup>-2</sup> ) | y = 4.24 + 1.48x                    | 7.62<br>(2)      | 0.022   |
| Citronella oil | 43.4 (37.4 – 51.7)                                                     | 199.0 (142.5 – 324.1)                                                   | y = 1.94x-3.17                      | 0.30<br>(2)      | 0.863   |
| Eucalyptus oil | 22.7 (20.3 – 25.2)                                                     | 63.8 (54.3 – 78.8)                                                      | y = 2.85x-3.87                      | 4.06<br>(2)      | 0.131   |
| Pine oil       | 23.2 (15.4 – 33.4)                                                     | 62.4 (41.0 – 176.2)                                                     | y = 2.98x-4.06                      | 4.35<br>(2)      | 0.113   |

higher than those of eucalyptus (22.7 and 63.8 ppm) and pine (23.2 and 62.4 ppm) oils, respectively (Table 4).

#### Field larvicidal efficacy of BEOs against larval *Culex quinquefasciatus*

In field larvicidal bioassays, all three essential oils individually or in combination produced 100% mortality in 3rd instar *Cx. quinquefasciatus* for 2 weeks which was significantly different from controls (One-way ANOVA week 1:  $df = 4, 16$ ,  $f$ -ratio = 3397.1,  $p = 0.0001$ ; week 2:  $df = 4, 16$ ,  $f$ -ratio = 2992.3,  $p = 0.0001$ ; Fig. 1). At the 3rd week, the combination of pine and eucalyptus oils maintained 100% mortality which was slightly reduced to  $91.3 \pm 3.2$  % in pine and  $94.0 \pm 1.5$  % in eucalyptus oils in comparison to  $75.0 \pm 5.0$  % mortality with citronella oil (One-way ANOVA:  $df = 4, 16$ ,  $f$ -ratio = 562.9,  $p = 0.0001$ ). At week 4, a significant higher mortality ( $96.0 \pm 1.0$  %) was observed in the combination of pine and eucalyptus oils compared to eucalyptus ( $88.3 \pm 0.3$  %), pine ( $82.3 \pm 1.3$  %) oils; but, the efficacy of citronella oil EC was significantly reduced to  $60.3 \pm 2.4$  % (One-way ANOVA:  $df = 4, 16$ ,  $f$ -ratio = 164.0,  $p = 0.0001$ ). Control showed overall 5–7.3% mortality.

#### Oviposition repellency of essential oils EC formulations under field conditions

Oviposition repellency test of BEOs formulations against *Cx. quinquefasciatus* under field conditions showed that the ovitraps treated with individual citronella, eucalyptus, and pine oils or the combination eucalyptus and pine oils did not receive any eggs in comparison to  $244.2 \pm 18.2$  eggs/trap in control during 1st week (One-way ANOVA:  $df = 4, 25$ ,  $f = 179.4$ ,  $p = 0.0001$ ; Fig. 2). At 2nd week, eucalyptus and pine treatments showed significantly less eggs in traps in comparison to citronella, and these were significantly lower than those of control (One-way ANOVA:  $df = 4, 25$ ,  $f = 338.1$ ,  $p = 0.0001$ ). Oviposition repellency in 3rd week was persistent in eucalyptus, pine, and a combination of both which were completely lost in citronella oil in comparison to control (One-way ANOVA:  $df = 4, 25$ ,  $f = 99.4$ ,  $p = 0.0001$ ).

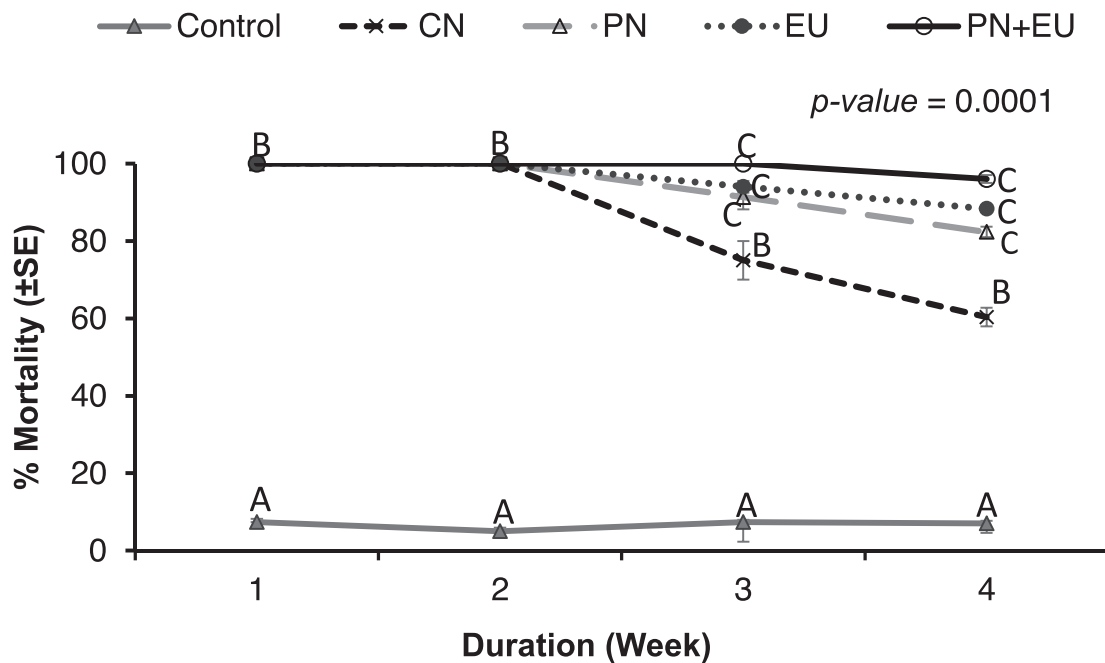
#### Estimation of oviposition attraction index (OAI)

The oviposition attraction indices of BEOs were mentioned in Table 5. All BEOs formulations showed complete repellency (-0.1) at 1st week. The treatment of eucalyptus, pine, and their combination at 2nd week resulted in high repellency (-0.9 to -0.1), respectively, which was slightly reduced at -0.5 at 3rd week in both treatments. However, citronella lost efficacy quickly and was -0.6 at the 2nd week and, then, reduced to 0.1 at the 3rd week (Table 5).

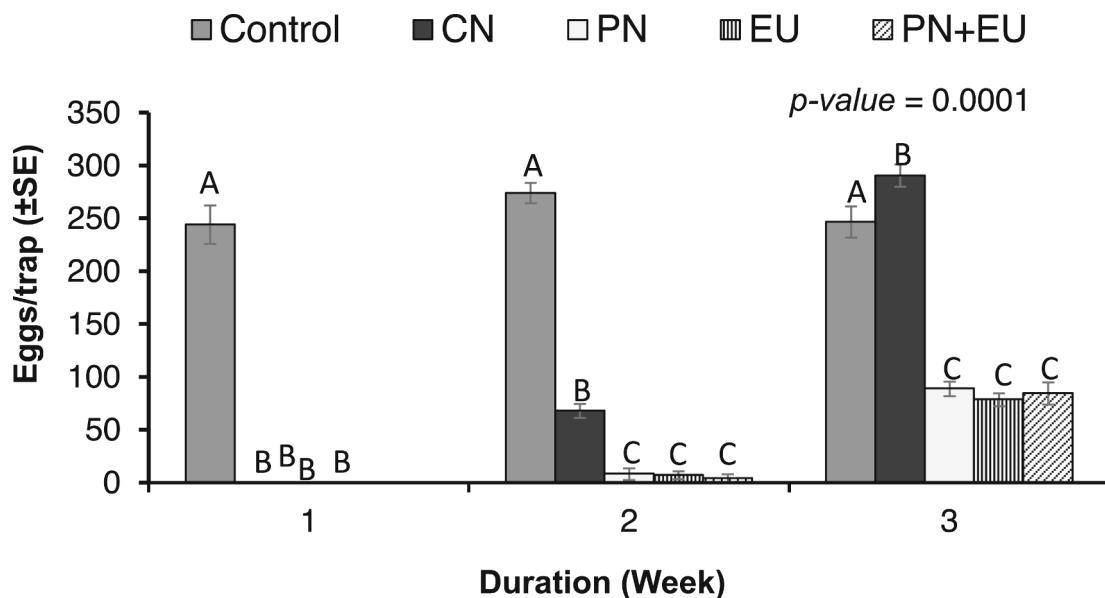
## Discussion

The management of larval mosquito population is a vital strategy to restrict the transmission of various vector-borne diseases (WHO, 2009). Synthetic organophosphate, temephos, is the most used larvicide globally which is categorized under higher hazardous level with a significantly higher recommended dose of 1 mg/L in comparison to insect growth regulators like pyriproxyfen, methoprene, diflubenzuron etc (WHO, 2009; Akiner and Eksi, 2015). The broadcast spray is commonly used to treat numerous larval habitats, but it has a tendency of unprecise delivery and also requires multiple repetitive applications to keep the mosquito population significantly low (Fonseca et al., 2013; Suman et al., 2014; Williams et al., 2014). This practice raises serious concerns over the safety of the environment and non-target organisms. In the present study, various emulsifiable formulations of citronella, eucalyptus, and pine BEOs were developed to achieve efficient larvicidal and oviposition repellent efficacy against the filarial vector, *Cx. quinquefasciatus*, surviving in urban and semi-urban areas. This study provides an ecofriendly vector management tool to kill the standing larval mosquito populations while making container habitats further immune to





**Fig. 1.** Field larvicidal efficacy different formulations of citronella (CN), eucalyptus (EU), pine (PN), and mix of eucalyptus and pine (EU + PN) essential oils against *Culex quinquefasciatus* larvae in 20 L water in plastic buckets treated with 300 ppm. Control was untreated water for larval bioassay in both experiments. One-way ANOVA was performed among various values of treatment and control on weekly basis. The values denoted with the same letters do not differ significantly ( $p < 0.05$ ).



**Fig. 2.** Oviposition repellency (eggs/trap) of emulsifiable concentrate formulations of citronella (CN), eucalyptus (EU), pine (PN) and combination of eucalyptus and pine (PN + EU) for *Culex quinquefasciatus* mosquitoes on weekly basis. Bars denoted with similar letters do not differ significantly on weekly basis (One-way ANOVA at  $p < 0.05$ ).

oviposition at the same time. These findings may be important for the development of a better eco-friendly management strategy “Kill and Push” against container inhabiting mosquitoes; where the “Kill” represents the killing of standing larval populations in the breeding habitats, and the “Push” represents the repelling of gravid females from the treated oviposition habitats.

Botanical essential oils are rich in chemical components eliciting positive responses for larvicidal and repellent activities in mosquitoes (Amer and Mehlhorn, 2006; Ansari et al., 2005; Andrade-Ochoa et al., 2018; Ghosh et al., 2012; Hsu et al., 2013). The present study on the

chemical characterization using the GC–MS method showed the presence of 17, 16, and 19 compounds in citronella, eucalyptus, and pine oils that provides 88.0%, 87.5%, and 93.0% of chemical compositions of these oils, respectively, which includes several known compounds such as Caryophyllene, Citronellol, Eugenol, Geraniol in citronella oil, (-)-trans-Pinane, 3-Carene,  $\alpha$ -Pinene, Camphene, Eucalyptol, O-Cymene in eucalyptus oil, and 3- and 4-Carene,  $\alpha$ -Pinene,  $\alpha$ -Terpineol,  $\gamma$ -Terpinene, Longifolene etc. in pine oil. Cheng et al. (2009) have extracted 12 pure constituents from *Eucalyptus camaldulensis* and *Eucalyptus urophylla* and also found six effective constituents, out of which  $\alpha$ -terpinene was

**Table 5**

Oviposition attraction index (OAI) of emulsifiable concentrate formulations of botanical essential oils (BEOs) [citronella, eucalyptus, pine, and combination of eucalyptus and pine (EU + PN)] for oviposition repellency of *Culex quinquefasciatus* mosquito under field evaluations.

|             | Oviposition attraction index   |      |      |
|-------------|--------------------------------|------|------|
|             | Post-treatment duration (Week) |      |      |
|             | 1                              | 2    | 3    |
| <b>BEOs</b> |                                |      |      |
| Citronella  | −1                             | −0.6 | 0.1  |
| Pine        | −1                             | −0.9 | −0.5 |
| Eucalyptus  | −1                             | −1.0 | −0.5 |
| EU + PN     | −1                             | −1.0 | −0.5 |

Note: The OAI response ranges between + 1 (maximum attractant) and −1 (maximum repellent), and 0 (neutral).

shown to be the best larvicidal agents against both *A. aegypti* and *A. albopictus* larvae. The promising larvicidal efficacy with pine and eucalyptus ECs ( $LC_{50}$  = 23.2 ppm and 22.7 ppm) in comparison to citronella EC ( $LC_{50}$  = 45.9 ppm) against larvae of *Cx. quinquefasciatus* confirms the mosquitocidal activities of these oils. Earlier studies on BEOs have reported lesser efficacy of essential oils, i.e., pine extract, *Pinus longifolia*  $LC_{50}$  was 85.7 ppm (Ansari et al., 2005). Sugumar et al. (2014) have developed nanoemulsion formulations with surfactants to improve the larvicidal efficacy of eucalyptus oil against *Cx. quinquefasciatus* mosquitoes, however, it required 200–250 ppm to achieve complete mortality of larvae. Similarly, different fractions of neem seed oil produced  $LC_{50}$  between 142.3 and 209.7 ppm against *Ae. albopictus* (Benelli et al., 2015). It shows that the formulated products can provide better larvicidal efficacy in comparison to isolates or fractions. Although, botanical essential oils are required higher doses to achieve the  $LC_{50}$  and  $LC_{90}$  in comparison to temephos, they are considered to be safe for the environment and non-target organisms and become an alternative option when temephos loses its efficacy.

Furthermore, the field larvicidal efficacy of a formulation is an important criteria to consider it appropriate for mosquito control applications. In our study, eucalyptus and pine oil EC formulations have shown promising larvicide activity for several weeks in field conditions. These ECs have produced 100% larvicidal effects against *Cx. quinquefasciatus* for 2 weeks at 300 ppm concentration that was prepared in 20 L water volume in a plastic bucket test. The efficacy was slightly reduced at week 3 and week 4. It was interesting to note that the combination of eucalyptus and pine oils enhanced the stability more than the individual eucalyptus and pine oils. Contrary to it, citronella oil lost the larvicidal efficacy soonest. These formulations can be used with conventional sprayers such as back-pack, handheld, or low-volume sprayers to treat a variety of larval breeding sites. By considering the larvicidal efficacy, stability and applications ease of these BEOs formulations, we suggest that eucalyptus and pine oils formulation can be used against container mosquitoes.

Additionally, oviposition repellents can be an important tool to secure mosquito breeding habitats from subsequent oviposition and result in the reduction of population in a similar manner of source reduction of mosquitoes or the barrier treatment for container mosquitoes. Essential oils have been well used as an adult repellent to avoid mosquito bites, but, fewer studies have been reported on oviposition repellency or deterrence (Benelli et al., 2015; Coria et al., 2008; Kempraj and Bhat, 2011; Maheswaran and Ignacimuthu, 2012; Xue et al., 2001; Cheah et al., 2013). Oviposition deterrents or repellents were produced short-term efficacy under field conditions. For example, neem seed oil has shown reduction in the number of *Ae. albopictus* eggs in a treated container more than the control for a week (Benelli et al., 2015). In search of effective oviposition deterrent, synthetic adult repellents such as DEET, DEPA, and DEB have also been evaluated for oviposition repellency (Tikar et al., 2014). Recently, *Artemisia verlotiorum* and *Ruta*

*chalepensis* essential oils have also shown oviposition deterrence against *Ae. albopictus* (Bedini et al., 2018). Similar to our findings for chemical composition of eucalyptus oil, da Silva et al. (2015) found almost similar amount of eucalyptol in *Commiphora leptophloeos*. In the present study, eucalyptus, pine, and a combination of eucalyptus and pine oil showed complete inhibition of oviposition in treated habitats resulting in −1.0 and −0.9 of OAI against *Cx. quinquefasciatus* for week 1 and 2. However, oviposition repellency of these oils was reduced to half at 3rd week. Citronella oil did not protect oviposition habitats after week 1 of treatment in comparison to other oils and received significant oviposition in treated habitats. The oviposition repellency in our study supports the findings of da Silva et al. (2015) who showed effective deterrent activity of *Commiphora leptophloeos* oil, (E)-Caryophyllene and  $\alpha$ -Humulene against the dengue vector, *Ae. aegypti* mosquitoes. The deterrence in treated cups was significantly different from control, but unable to protect habitat completely at 100 ppm in the study of da Silva et al. (2015). In our study, the promising oviposition repellent activity was shown in eucalyptus and pine oil treatments at 300 ppm during week 1 and 2. The use of oviposition repellents may supplement the sanitation process necessary for the reduction of mosquito populations in infested areas (Unlu et al., 2013), particularly in active key breeding sites of mosquitoes which are unable to be removed or cleaned (WHO, 2009). Therefore, the EC formulations of eucalyptus, pine or their combination can be a cost-effective and eco-friendly treatment in comparison to the removal or sanitization of breeding habitats manually.

Moreover, botanical essential oils may have another advantage over many synthetic larvicides in insecticide resistance management. The long-term and injudicious use of insecticides have resulted in the development of resistance or tolerance in various mosquitoes that have led to the failure of vector control programs globally (Marcombe et al., 2014; Brogdon and McAllister, 1998). Several essential oils such as clove, geranium, oregano, cinnamon, and basil plant have shown synergistic effects similar to PBO and have increased susceptibility of resistant mosquitoes to different insecticides, i.e., *Ae. aegypti* with kdr mutation resistant to deltamethrin and permethrin, *An. gambiae* resistant to carbamate and organophosphate, and *Ae. albopictus* resistant to organochlorine (Elia-Amira et al., 2018; Gross et al., 2017; Norris et al., 2019). Interestingly, the  $LC_{90}$  of field populations of *Cx. quinquefasciatus* (0.01 ppm) was closer to WHO diagnostic dose (0.012 ppm) suggesting that the mosquito population may have resistance to temephos in near future. Therefore, it may be useful to perform further studies for collecting data on the synergistic effects of these essential oils to formulate a strategy of insecticide resistance management in vectors.

## Conclusion

The present study established the larvicidal and oviposition repellent efficacy of emulsifiable concentrate formulations developed from citronella, eucalyptus, and pine oils against *Cx. quinquefasciatus* populations having lower insecticide resistance status against temephos. The combination of both eucalyptus and pine over individual oils showed an advantage in the enhancement of larvicidal efficacy under field conditions. The oviposition repellency test revealed that pine and eucalyptus oils as individual and, in a combination, protected mosquito breeding sites from further oviposition for 3 weeks. Citronella oil was the weakest larvicide and oviposition repellent against filaria mosquitoes. The chemical profiling of these essential oils provides information to develop new formulations with various chemical compounds present in BEOs. The use of BEOs can be significant in reduction of synthetic poisonous insecticides load in peridomestic habitats used for mosquito management. Based on promising higher larvicide and oviposition repellent efficacies, this study also suggests that eucalyptus, pine, and their combinations have the potential for the development of eco-friendly “Kill and Push concept” against the mosquitoes.

## Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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